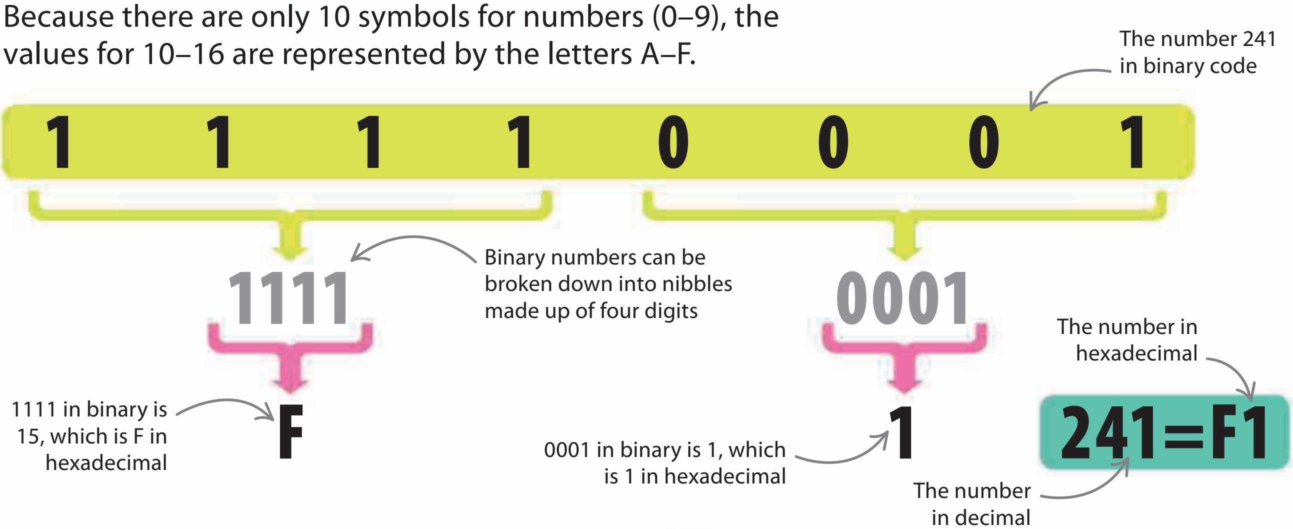
Hexadecimal

When using numbers in computer programs, a base of 16 is often used because it's easy to translate from binary. Because there are only 10 symbols for numbers (0–9), the values for 10–16 are represented by the letters A–F.

▽ **Understanding nibbles**
A "nibble" is made up of four binary digits, which can be represented by one hexadecimal digit.



▽ **Comparing base systems**
Using this table, you can see that expressing numbers in hexadecimal gives the most information with the fewest digits.

DIFFERENT BASES

Decimal	Binary	Hexadecimal
0	0 0 0 0	0
1	0 0 0 1	1
2	0 0 1 0	2
3	0 0 1 1	3
4	0 1 0 0	4
5	0 1 0 1	5
6	0 1 1 0	6
7	0 1 1 1	7
8	1 0 0 0	8
9	1 0 0 1	9
10	1 0 1 0	A
11	1 0 1 1	B
12	1 1 0 0	C
13	1 1 0 1	D
14	1 1 1 0	E
15	1 1 1 1	F

REMEMBER

Bits, nibbles, and bytes

A binary digit is known as a "bit", and is the smallest unit of memory in computing. Bits are combined to make "nibbles" and "bytes". A kilobit is 1,024 bits. A megabit is 1,024 kilobits.

Bits: Each bit is a single binary digit—a 1 or 0.

Nibbles: Four bits make up a nibble—enough for one hexadecimal digit.

Bytes: Eight bits, or two hexadecimal digits, make up a byte. This gives us a range of values from 0 to 255 (00 to FF).